

Bank Marketing Campaign Analysis Report

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Dataset URL: <https://www.kaggle.com/janiobachmann/bank-marketing-dataset>

Github Repository URL: <https://github.com/Ravindu62/bis-4116-assignment.git>

1. Executive Summary

This report presents an analysis of direct marketing campaigns (phone calls) of a Portuguese banking institution. The classification goal is to predict if the client will subscribe to a term deposit (variable `deposit`).

Key Findings:

- **Call Duration is Critical:** There is a strong positive correlation between call duration and subscription success. Longer engagement typically leads to a positive outcome.
- **Account Balance Matters:** Clients who subscribed had, on average, a 40% higher account balance (€1,804) compared to those who did not (€1,280).
- **Campaign Fatigue:** Repeated contacts during the same campaign (`campaign` variable) show a slight negative correlation with success, suggesting that pestering clients is ineffective.
- **Dataset Balance:** The dataset is surprisingly well-balanced (47.38% subscription rate), making it excellent for training classification models without needing heavy resampling.

2. Business Domain & Problem Statement

2.1 Context

In the banking industry, term deposits are a major revenue source. Marketing campaigns, often telephonic, are the primary tool to sell these products. However, these campaigns can be costly and inefficient if not targeted correctly.

2.2 Business Goal

The objective is to identify the characteristics of customers who are more likely to subscribe to a term deposit. This insight will allow the bank to:

- Optimize resource allocation (focus on high-probability leads).
- Improve call scripts (understanding the importance of duration).
- Reduce customer annoyance (avoiding over-calling).

2.3 Dataset Overview

The dataset contains **11,162** records with 17 features, including:

- **Client Data:** Age, Job, Marital Status, Education, Balance, Housing/Personal Loan.
- **Campaign Data:** Contact communication type, Last contact day/month, Duration, Number of contacts.
- **History:** Days since last campaign (`pdays`), Previous contacts (`previous`), Outcome of previous campaign (`poutcome`).

3. Data Preprocessing & Cleaning

3.1 Inspection

The dataset was loaded and inspected for data integrity.

- **Total Records:** 11,162
- **Missing Values:** 0 found.
- **Duplicates:** 0 found.

The data quality is high, requiring no imputation or row dropping for this analysis.

4. Exploratory Data Analysis (EDA)

4.1 Target Variable Distribution

The dataset shows a relatively balanced distribution between those who subscribed ('yes') and those who did not ('no').

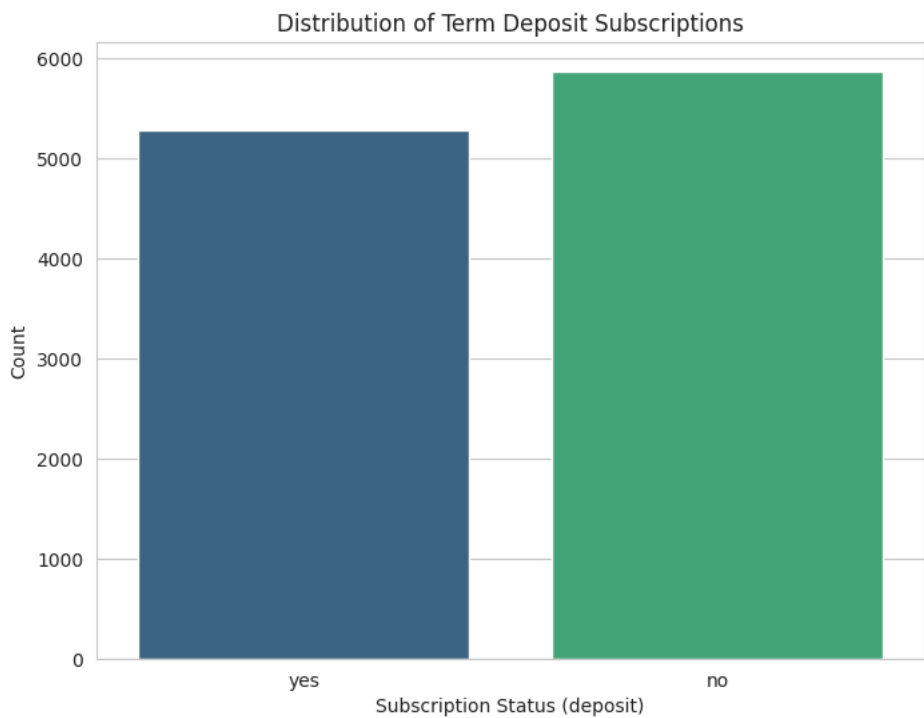


Figure 1: Distribution of Term Deposit Subscriptions.

4.2 Age Distribution

The age distribution is similar for both groups, with a slight peak in the 30-40 range. However, the density of subscriptions is slightly higher in younger (<30) and older (>60) demographics, suggesting these groups might be more receptive targets than the middle-aged working population.

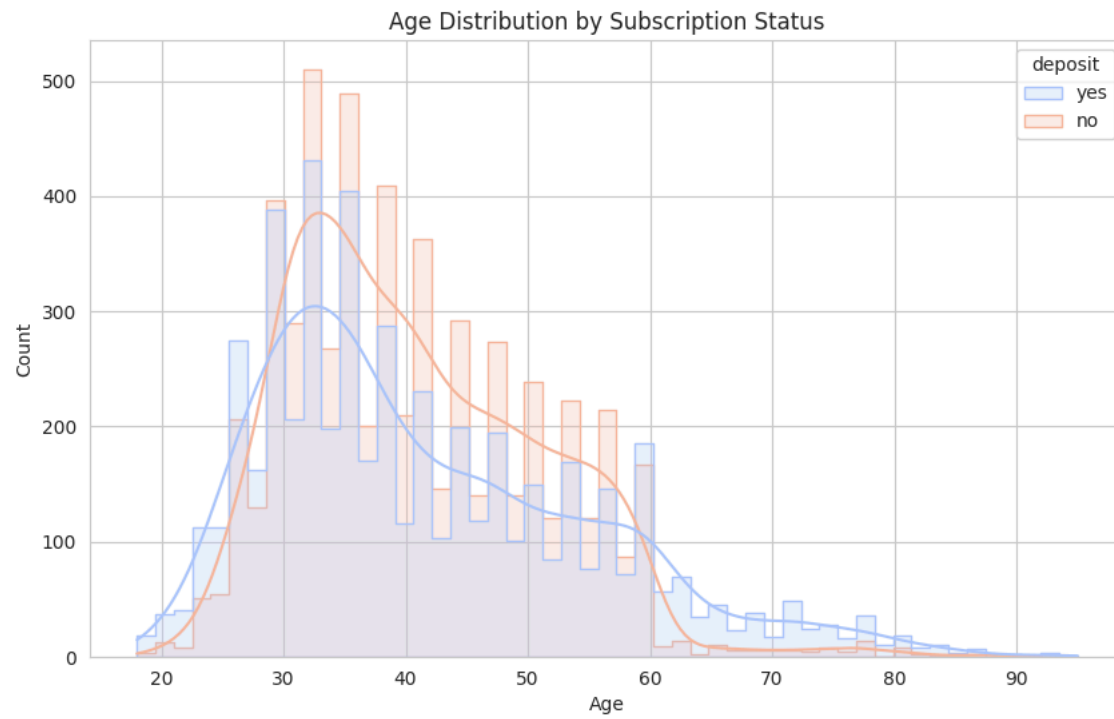


Figure 2: Age Distribution by Subscription Status.

4.3 Job Type Impact

Certain job categories show higher engagement. Students and retired individuals, despite having lower raw numbers, often have higher subscription *rates* compared to blue-collar workers.

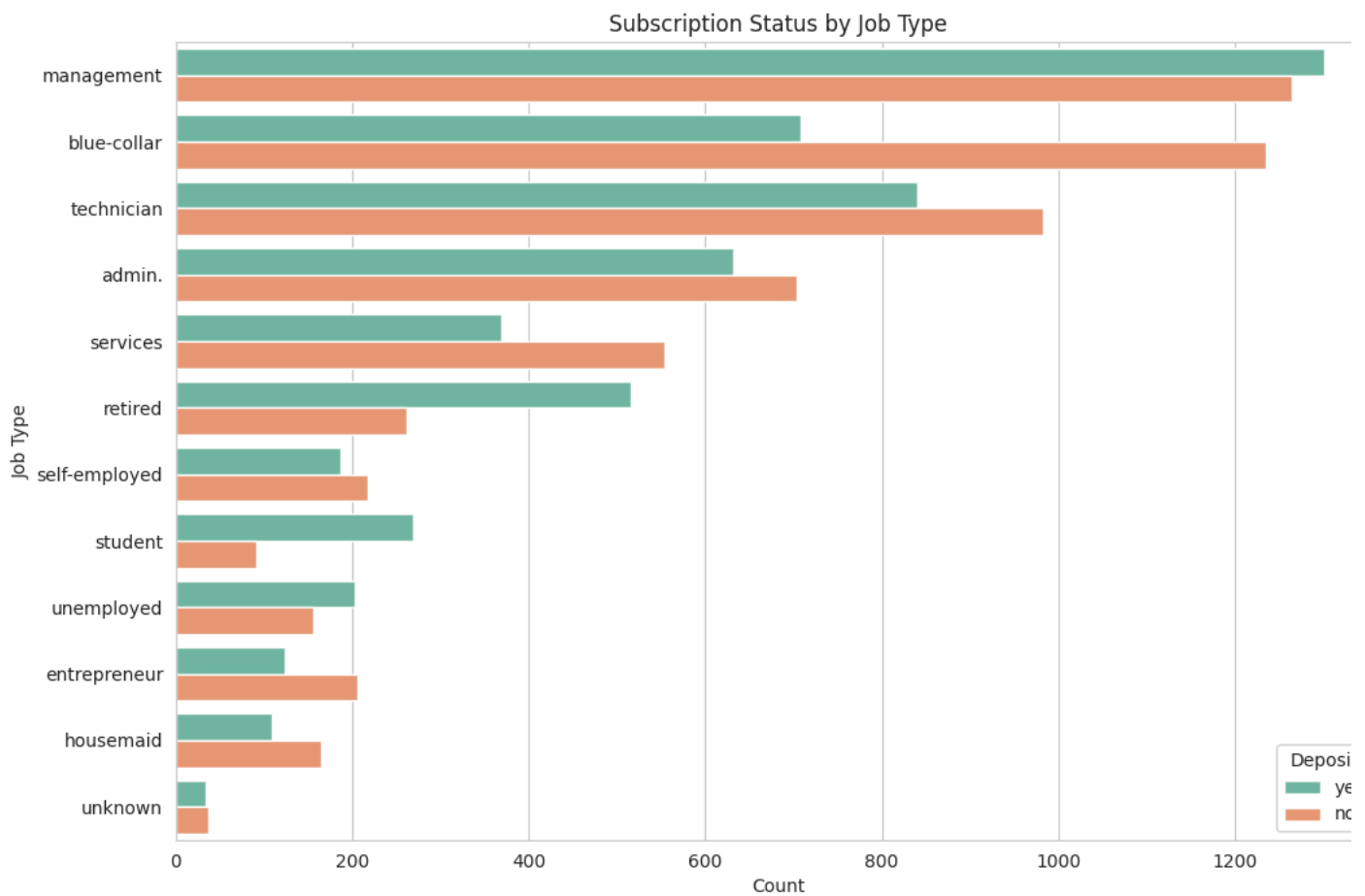


Figure 3: Subscription Status by Job Type.

4.4 Financial Status (Balance)

I observed a clear distinction in financial health. Subscribers generally possess higher average balances.

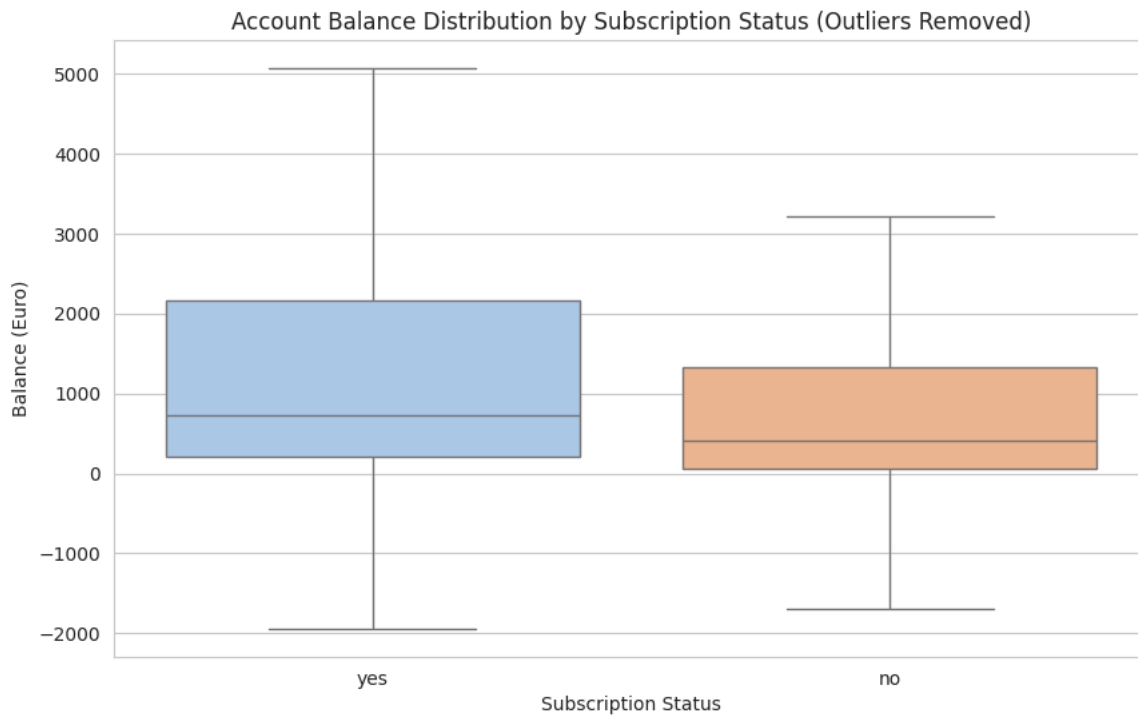


Figure 4: Account Balance Distribution (Outliers removed for clarity).

Statistic:

- Mean Balance (Subscribed): **€1,804**
- Mean Balance (Not Subscribed): **€1,280**

5. Statistical Analysis & Correlations

5.1 Correlation Matrix

I analyzed the correlation between numerical features and the target variable.

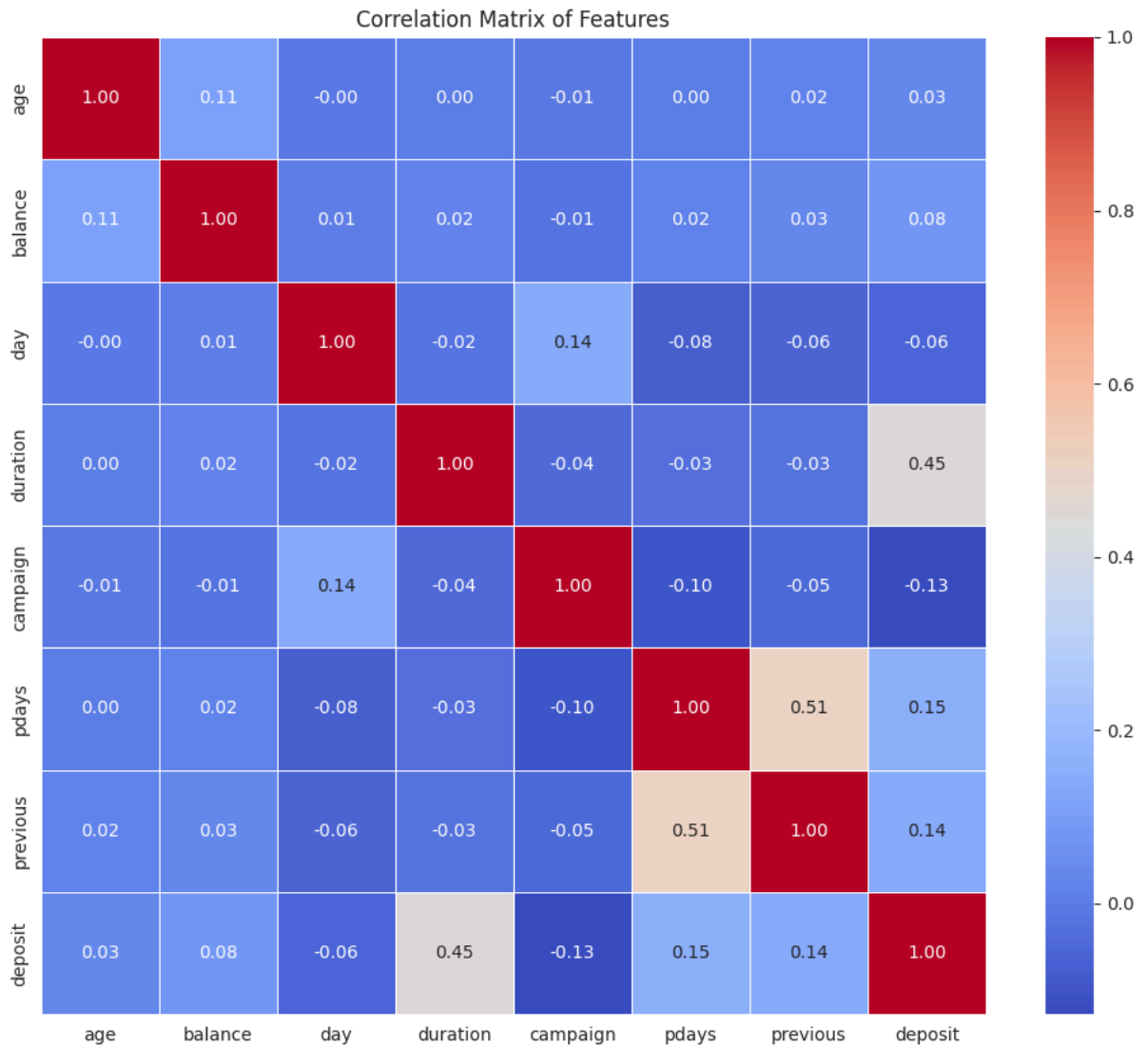


Figure 5: Correlation Matrix of Numeric Features.

5.2 Key Drivers

1. **Duration (0.45):** The strongest predictor. This is intuitive; a longer conversation implies interest. However, for predictive modeling, this variable should be used with caution as the duration is not known *before* the call is made.
2. **Previous Contact (0.14):** Successful previous interactions increase the likelihood of current success.
3. **Campaign (-0.13):** A negative correlation indicates that more calls to the same client during this campaign decrease the probability of subscription.

6. Conclusions & Recommendations

6.1 Strategic Recommendations

1. **Focus on Engagement:** Since duration is the top predictor, train agents to engage customers in meaningful conversation rather than rushing through scripts.
2. **Target High-Balance Segments:** Prioritize customers with higher account balances, as they have the liquidity to invest in term deposits.

3. **Respect Contact Limits:** Limit the number of calls per campaign (`campaign` variable). If a customer hasn't converted after the first few calls, further attempts yield diminishing returns and may harm the relationship.
4. **Demographic Targeting:** tailored campaigns for "Students" and "Retired" individuals could yield high conversion rates given their favorable response patterns.